

Pre-course preparation

Efficient Bayesian Analysis of Longitudinal and Survival Data Using R-INLA
ISCB 2026, Freiburg, Germany

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Before the course

Running the code during the course is **optional**. If you want to run the code, please complete the following steps **before the course**. Installing packages can take time, so do not leave this for the last minute.

GITHUB : <https://github.com/DenisRustand/ISCB2026>

R version

You need **R** ≥ 4.6 . Check with:

```
R.version.string
```

If your version is older, update from <https://cloud.r-project.org>.

Package installation

```
install.packages("INLA",  
  repos = c(getOption("repos"),  
    INLA = "https://inla.r-inla-download.org/R/testing"))  
  
install.packages(c("INLAjoint", "ggplot2", "ggpubr", "survival",  
  "mvtnorm", "PermAlgo", "JM",  
  "rmarkdown", "knitr", "sn"))
```

Verify installation

Run the following to check that the core packages load and compute correctly:

```
library(INLA)  
library(INLAjoint)  
cat("INLA version:", inla.version()$version, "\n")  
cat("INLAjoint version:", as.character(packageVersion("INLAjoint")), "\n")  
  
# Toy example: intercept of  $y \sim 1$  with two observations (2 and 8) should be 5  
m <- inla(y ~ 1, data = list(y = c(2, 8)),  
  control.compute = list(config = TRUE))  
cat("Intercept (expected 5):", round(m$summary.fixed["(Intercept)", "mean"], 1), "\n")  
invisible(inla.posterior.sample(1, m))  
cat("Posterior sampling: OK\n")
```

On Linux, if INLA throws a binary error, try switching binaries:

```
inla.binary.install()
```

Common issues

- **Linux users:** you may need system libraries `libudunits2-dev`, `libgdal-dev`, `libgeos-dev`, `libproj-dev`, `libsqlite3-dev` and `libgs1-dev`.
- **Mac users:** make sure Xcode command line tools are installed (`xcode-select --install`)
- **Windows users:** Rtools may be required for some packages